

Puneet Velidi

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EDUCATION

University of Victoria

MSc in Statistics (thesis-based, fully-funded)

Co-supervisors: Prof. Farouk Nathoo & Prof. Michelle Miranda

Committee members: Tanya Garcia, Maryclare Griffin, and Enrico Amico | External examiner: Jörn Diedrichsen

Thesis: *Methods for Censored Covariates, Spatial Transcriptomic Data and Neural Manifolds*. [Oral examination](#)

Victoria, BC

Jan. 2025 – Dec. 2026

Cornell University

BA in Statistics & Economics

Ithaca, NY

Sep. 2018 – Dec. 2022

PUBLICATIONS (†indicates joint first authorship)

Puneet Velidi, Maryclare Griffin, Juxin Liu, Brian Richardson, Jing Qian, Roland Matsouaka, Sujit Ghosh, Tanya Garcia, Farouk Nathoo. *SPARTACCUS: Semiparametric Bayesian Proportional Hazards Model with a Censored Covariate in Huntington's Disease*. **In Preparation**, 2026.

Puneet Velidi, Zhengxiao Wei, Farouk Nathoo. *Covariance Nonstationarity is Evident in Spatial Transcriptomics and Provides a New Categorization of Spatially Varying Genes*. **Submitted for publication**, 2026. [bioRxiv](#)

Puneet Velidi, Michelle Miranda, Farouk Nathoo, Ashery Mbilinyi, Cédric Beaulac. *Shape Analysis as an Inductive Bias for Glioma Grading from Tumor Contours*. **Submitted for publication**, 2026. [arXiv](#)

Puneet Velidi, Enrico Amico, Farouk Nathoo, Michelle Miranda. *Neural manifold connectomics reveals multiregime functional connectivity*. **Under review**, 2026. [bioRxiv](#)

Fabio Luchese, **Puneet Velidi**, . . . , Paolo Cassano. *Transcranial Photobiomodulation on Language and Cognitive Performance in Down Syndrome: A Pilot Randomized Sham-Controlled Trial*. **Under review**, 2026. [medRxiv](#)

B. Ferreras, . . . , **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation in Bipolar Disorder: A Case Series on Executive Skills and fMRI BOLD Signal*. **Submitted for publication**, 2026. [SSRN](#)

Guillermo Gonzalez-Garibay, **Puneet Velidi**, Akila Weerasekera, Nikos Makris, Carlos Lohmann, Katherine Anne Collins, Naomi Gaggi, Dan Vlad Iosifescu, Paolo Cassano. *Volumetric Changes Induced by Transcranial Photobiomodulation Therapy in Major Depressive Disorder: An Exploratory Neuroimaging Study*. **Submitted for publication**, 2026. [SSRN preprint](#)

Puneet Velidi[†], Zhengxiao Wei[†], Shreena N. Kalaria, Yimeng Liu, Céline M. Laumont, Brad H. Nelson, Farouk S. Nathoo. *Generalized Jeffreys's Approximate Objective Bayes Factor: Model Selection Consistency, Finite Sample Accuracy, and the Statistical Evidence in 71,126 Clinical Trial Findings*. **Under revision**, 2025. [arXiv](#)

S. Parker Singleton[†], **Puneet Velidi**[†], Louisa Schilling, Andrea I. Luppi, Keith Jamison, Linden Parkes, Amy Kuceyeski. *Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory*. **Biological Psychiatry: Cognitive Neuroscience and Neuroimaging**, 2024. [Link](#)

SCHOLARSHIPS, AWARDS, RECOGNITIONS

2026	ICSA-Canada Chapter — Student/Postdoc Travel Scholarship Award This award supports travel expenses for the International Chinese Statistical Association – Canada Chapter 2026 Symposium in Montreal.	600 CAD
2025	Labs4 — Technology Readiness Level-Up (TRL [^]) Fellowship Lab-to-market grant; ChimeraML, combinatorial nanobody design with an ensemble of protein ML models for immune checkpoint inhibitor therapy. Funded by Mitacs, New Ventures BC, and BCIT. Labs4 TRL [^] Pitch Competition, BC Region, 2nd place.	15,000 CAD
2025	University of Victoria — Betty & Gilbert Kennedy Graduate Scholarship in Math. & Stats. Awarded to academically outstanding students in the Department of Mathematics and Statistics upon the recommendation by the department (<i>Donor Award</i>).	5,130 CAD
2025	Maud Menten Institute/PIMS — Research Accelerator Award This award supports a collaboration with the MGH Photobiomodulation Lab on Bayesian analysis of the TransPhoM-DS clinical trial.	5,000 CAD

2025	University of Victoria — FGS Graduate Fellowship (Master's) This fellowship is awarded to students of high academic standing upon recommendation by their department. <i>The typical M&S FGS Master's Fellowship is 2,500 CAD.</i>	<i>5,000 CAD</i>
2025	University of Victoria — FGS Travel Award & CUPE Conference Award These awards supported travel expenses for CNS 2025.	<i>1,050 CAD</i>
2025	3MC–PIMS–IDMS–ICMS — Summer School Supported Participant This recognition supported travel expenses + room and board for the two-week 2025 3MC–PIMS–IDMS–ICMS Summer School.	—
2025	Maud Menten Institute/PIMS — MMI HQP Summit Supported Participant This recognition supported travel expenses + room and board for the 2025 MMI HQP Summit.	—

COMPETITIONS

2026	Y Combinator (P26) Selected for an in-person interview in San Francisco out of 80k applications (approx. top 2.5%).	—
2026	Critical Ops: A DC National Security Hackathon (spon. OpenAI, Meta, Squadra Ventures) 1st place out of 166 participants , ENEL (drone swarm 3D Gaussian Splat). Won demo opportunities with OpenAI and Meta .	—
2026	SCSP AI+ Expo Hackathon 2nd place out of 200+ participants , 1st place DC region, Electrical Grid Track. Franklin, IoT grid sound anomaly detection & agent-to-agent load negotiation.	<i>5,000 USD</i>
2026	Stanford University RNA 3D Folding Competition Placed in the top 3% of 1,867 teams; Silver Medal .	—

PRESENTATIONS

Oral Presentations

Brain Fingerprinting of Principal-component derived fMRI BOLD functional connectivity. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC. Invited Talk, **Invited Session Chair**.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 9th International Conference on Econometrics and Statistics, August **2026**. Kyoto, JP. Co-author of invited talk.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON. Contributed Talk.

“Applications of Protein Folding Models in Immunotherapeutic Research.” UBC Math Bio Seminar, January **2026**. Invited Seminar.

“70,000 Leagues Under the Sea of p -Values.” Maud Menten Institute / Mathematical and Statistical Biology Seminar, October **2025**. Invited Seminar.

“70,000 Leagues Under the Sea of p -Values.” UBC Stats Seminar, October **2025**. Invited Seminar.

Statistical methods for censored covariates. 2025 WNAR/IMS Annual Meeting, June **2025**. Whistler, BC. Co-author of discussant remarks for invited session.

“Network Control Theory in Neuroscience: Characterizing the Brain’s Energy Landscape.” Cognition and Brain Sciences Seminar, University of Victoria, March **2025**. Invited Seminar.

“Network Control Theory in Neuroscience: Characterizing the Brain’s Energy Landscape.” Maud Menten Institute / Mathematical and Statistical Biology Seminar, University of Victoria, February **2025**.

Poster Presentations

Acute Effects of Transcranial Photobiomodulation On Glymphatic CSF Dynamics in Two Patients with Amnesic Mild Cognitive Impairment or Early Alzheimer Disease. Clinical Research Day, Mass General Brigham, October 30, **2026**. Virtual poster presentation (upcoming).

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC.

Bayesian Analysis of the TransPhoM-DS Clinical Trial. Organization for Human Brain Mapping 2026 Annual Meeting, June **2026**. Bordeaux, FR.

Ahab's Harpoon: Calibrated irreproducibility detection with approximate objective Bayes factors. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON.

Reassessing the Statistical Evidence in Clinical Trials with Extended Approximate Objective Bayes Factors. Maud Menten Institute 2025 HQP Summit and Working Group Meetings, July **2025**. Edmonton, AB.

Photobiomodulation therapy-induced volumetric brain changes on patients with depression. Cognitive Neuroscience Society 2025 Annual Meeting, March **2025**. Boston, MA.

Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory. Organization for Human Brain Mapping 2024 Annual Meeting, June **2024**. Seoul, SK.

Network control analysis of individuals with substance use disorder. Organization for Human Brain Mapping 2023 Annual Meeting, July **2023**. Montreal, QC.

RESEARCH EXPERIENCE

Graduate Researcher

Victoria, BC

University of Victoria — Department of Mathematics and Statistics

Jan. 2025 – Present

Nathoo Lab, PI: Prof. Farouk Nathoo & Prof. Michelle Miranda

Visiting Researcher

Birmingham, UK

University of Birmingham — School of Mathematics

Aug. 2025

AMICO Lab, PI: Prof. Enrico Amico

Postbacc Researcher

Boston, MA

Harvard Medical School & MGH — Division of Neuropsychiatry and Neuromodulation

May 2024 – Dec. 2024

MGH Photobiomodulation Laboratory, PI: Prof. Paolo Cassano

- Statistical analysis of TransPhoM-DS, TRIADE, TPEB, TRAP-AD clinical trials.

Undergraduate Researcher

Ithaca, NY

Weill Cornell Medicine — Department of Radiology

Jan. 2022 – Dec. 2022

Computational Connectomics Laboratory, PI: Prof. Amy Kuceyeski

PROFESSIONAL EXPERIENCE

Founder

Vancouver, BC & Washington, DC

ChimeraML — TRL Level-Up (Labs4) Fall 2025 Cohort

Sept. 2025 – Present

- Combinatorial nanobody design using an ensemble of protein ML models for immune checkpoint inhibitor therapy.
White paper available upon request.

Software Engineer II

Bentonville, AR

Walmart Global Tech — EBS

Feb. 2023 – May 2024

- *Early Career Recognition*, awarded to <10% of 2023 Walmart Global Tech Development Program.

AI/ML & NextGen Analytics Intern

New York, NY

Memorial Sloan Kettering Cancer Center — DigITs

June 2022 – Aug. 2022

TEACHING EXPERIENCE

Teaching Assistant

Victoria, BC

University of Victoria — Department of Mathematics and Statistics

Jan. 2025 – April 2025

Math Support Center Tutor

Ithaca, NY

Cornell University — Department of Mathematics

Sep. 2021 – Dec. 2022

WORKSHOPS AND SHORT COURSES

BIRS Workshop: A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments. *Banff International Research Station (BIRS), Banff, AB.* May 10–15, **2026**.

Five-day invited workshop on 2- and 3-D mechanobiochemical models of single and collective cell migration.

eScience JASP Hackathon. *Netherlands eScience Center, Amsterdam, NL.* February 16–17, **2026**.

Developed a JASP module alongside the core JASP programming team.

3MC-PIMS-IDMS-ICMS Summer School on Quantitative Molecular and Cellular Biology. *University of Manitoba, Winnipeg, MB.* June 16–27, **2025**.

Two-week school on quantitative methods for structural, molecular, and cellular biology, using cancer as an integrative example.

TECHNICAL REPORTS

Computational Design of Cross-Reactive Nanobodies Targeting the Human Chemokine Receptor Family. ChimeraML, **2026**. Available upon request.

Immunodynamic Modeling in Melanoma and Gastric Cancer. Supervised by Kang-Ling Liao and Michael Chan.

3MC-PIMS-IDMS-ICMS Summer School on Quantitative Molecular and Cellular Biology, **2025**.

PROFESSIONAL SERVICE

Network Neuroscience, Reviewer

Human Brain Mapping, Reviewer (with Supervisor)

JASA Applications and Case Studies, Reviewer (with Supervisor)